Advanced control (LTI part)

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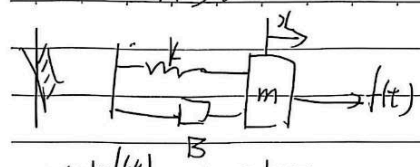
Advaced控制理论



$u(t)=f(t)$, $x_c(t)=x$

$m\ddot{x}+B\dot{x}+kx=f(t)$

$ms^2X(s)+Bs\cancel{X}X(s)+\cancel{k}kX(s)=F(s)$

$z_1=x,\ z_2=\dot{x}$ 则 $\dot{z}_1=z_2$, $\dot{z}_2=\ddot{x}$

则 $\begin{bmatrix}\dot{z}_1\\ \dot{z}_2\end{bmatrix}=\begin{bmatrix}0 & 1\\ -\frac{k}{m} & -\frac{B}{m}\end{bmatrix}\begin{bmatrix}z_1\\ z_2\end{bmatrix}+\begin{bmatrix}0\\ \frac{1}{m}\end{bmatrix}[u(t)]$

$y=[1\ 0]\begin{bmatrix}z_1\\ z_2\end{bmatrix}+[0][u(t)]$

若 $\dot{z}=Az+Bu$, $y=Cz+Du$ $\Longleftrightarrow$ $G(s)=\frac{X(s)}{R(s)}=\frac{Y(s)}{U(s)}=\frac{1}{ms^2+Bs+k}$

由 $\mathcal{L}\dot{z}=\mathcal{L}Az+Bu$

$sZ(s)=AZ(s)+BU(s)$

$Z(s)=[sI-A]^{-1}BU(s)$

$\mathcal{L}y=\mathcal{L}Cz+Du$

$Y(s)=CZ(s)+DU(s)$

$=[C(sI-A)^{-1}B+D]U(s)$

则 $G(s)=C(sI-A)^{-1}B+D$

A的特征值为G(s)的极点

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$\dot{x}_i(t) = a x_i(t)$

$\Rightarrow x_i(t) = x_i(0) e^{at}$

$\begin{cases} \dot{x}_1(t) = x_1(t) \\ \dot{x}_2(t) = -2x_2(t) \end{cases} \Rightarrow \begin{cases} x_1(t) = x_1(0) e^{t} \\ x_2(t) = x_2(0) e^{-2t} \end{cases}$

则 $\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$

$\frac{de^{At}}{dt} = Ae^{At}$

$e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & e^{\lambda_2 t} & \\ & & \ddots \\ & & & e^{\lambda_n t} \end{bmatrix}$

由 $A = P\Lambda P^{-1} \Rightarrow A^n = P\Lambda^n P^{-1}$

则由 $P^{-1}e^{At}P = I + \Lambda t + \frac{1}{2!}(\Lambda t)^2 + \cdots = e^{\Lambda t}$

则 $e^{At} = Pe^{\Lambda t}P^{-1}$ 解决

由 $e^{-At}\frac{d\vec{x}(t)}{dt} = e^{-At}(A\vec{x}(t) + B\vec{u}(t))$

则

$\frac{d(e^{-At}\vec{x}(t))}{dt} = \int_{t_0}^{t} e^{-A\tau} B\vec{u}(\tau)$

则 $e^{-At}\vec{x}(t)\Big|_{t_0}^{t} = \int_0^t e^{-A\tau} B\vec{u}(\tau)d\tau$

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则 $\vec{x}(t) = e^{A(t-t_0)}\vec{x}(t_0) + \int_{t_0}^{t} e^{A(t-\tau)} B\vec{u}(\tau)d\tau$

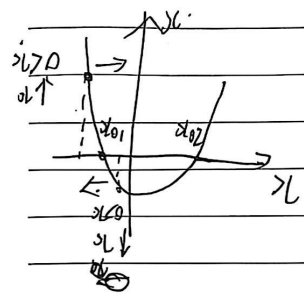
$= \Phi(t-t_0)\vec{x}(t_0) + \int_{t_0}^{t} \Phi(t-\tau) B\vec{u}(\tau)d\tau$

x_{01}, x_{02}

平衡点: Fixed Point ; Equilibrium Point

x_{01} 左右两侧则向 x_{01} 稳定移动, x_{01} stable

同理: x_{02} unstable



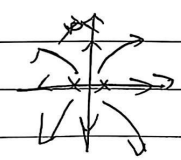
$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

令 $b = c = 0$. 则 $\begin{cases} \dot{x}_1 = ax_1 \\ \dot{x}_2 = dx_2 \end{cases} \Rightarrow$ Fixed Point $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

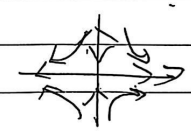
case 1: $a = d > 0$; $a > d > 0$

case 2: $a > 0$, $d < 0$

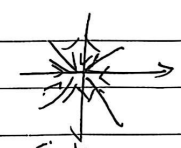
~~x~~ case 3: $a < 0$, $d < 0$



source
unstable



saddle



sink
stable

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General Form:

$\dot{x}=Ax$ 令 $x=Py$ $P=[v_1 v_2]$, $A=PAP\Lambda$

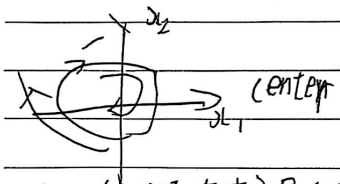
$\dot{y}=\Lambda y=\begin{bmatrix}\lambda_1 & \\ 0 & \lambda_2\end{bmatrix}y$

由 $\dot{y}=\Lambda y$ 可得相图

由 $x=Py$ 进行变化即得到 x 坐标系的相图

$G(s)=C(SI-A)^{-1}B+B$

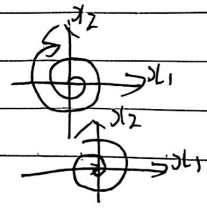
$(SI-A)^{-1}=\frac{(SI-A)^*}{|SI-A|}$ 故 $|I-A|=0$ 即为对应极点



特征值实部决定最终稳定与否

虚部决定稳定速度与振荡频率

Sum	$\lambda_{1,2}$	
Real	$\lambda_{1,2}<0$	stable
	$\lambda_1\lambda_2<0$	saddle
	$\lambda_{1,2}>0$	unstable
Complex $\lambda=a+bi$	$a=0$	center
	$a>0$	unstable
	$a<0$	stable



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$T\to 0$, $y[k]\to y(t)$

T 太长混叠 $\Rightarrow$ Shannon定理: $w_s>2w_n$

采样T > 计算T

对应OH: $\vec{x}[k+1]=e^{AT}\vec{x}[k]+\int_0^T e^{At}dt\,Bu[t]$

$=F\vec{x}[k]+Gu[k]$

$\lambda_i<0$ 则 v_i 指的随时间后会远离

$x_{k+1}=Ax_k+Bu_k$

$x_n=A^{n-1}Bu_0+\cdots+Bu_{n-1}=C_0U$

只有唯解: $Rank(C_0)=n$ 即可控系统, $x_0\xrightarrow{可控}x_n$

可控: 点 $\to$ 点 V, 轨迹 X

C_0: Controllability 可控性

对离散/连续而言的LTI均成立

$\lambda=a+bi$

stable - Lyapunov	$a\le 0$	LTI
渐近 - stable	$a<0$	
unstable	$a>0$	

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NonLinear System

Lyapunov 2nd Method:

$\dot{x} = f(x)$, $x=0$ 为平衡点

if (i) $V(0)=0$

(ii) $V(x) \geq 0$ in $D-\{0\}$, PSD

(iii) $\dot{V}(x) \leq 0$ in $D-\{0\}$ NSD

$\Rightarrow x=0$ stable

if (ii) (iii) 等号不成立则为渐近稳定性

即(ii) PD (iii) ND

引入u, Open loop $\Rightarrow$ closed loop

$\dot{x} = Ax + Bu$

特殊: $u=-kx$, ~~$k=[k_1 \; k_2]$~~

则 $\dot{x} = (A-Bk)x = A_{cl}x$

~~由Acl~~ 由自己设定的λ得 A_{cl}

LQR

λ的设计原则: Optimal Control (Cost Function)

① λ为虚数 $\lambda = \omega bi$ $J = \int_0^\infty (x^TQx + u^TRu)dt$

② $\Rightarrow$ 振动 Min(J)

③ λ决定收敛速度

④ 考虑u

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But $u=-kx \neq$ stable

考虑 $C_0 = [B \;\; AB \cdots]$ 得 $R(C_0)$ 得 controllability

C_0 决定有无可能设计λ控制器 控制 ~~Close~~ 使系统stable

选取k $\longrightarrow$ 改变 A_{cl} 的 λ_i $\xrightarrow{\text{控制}}$ 系统表现

Cost Funtion: $J = \int_0^\infty (x^TQx + u^TRu)dt$

Penalty: $x/u \neq 0$ 时的惩罚

R个, u对J的影响小, 使J小则u↓, 控制需耗力↓

$x_i \neq 0$ 时 J个, 确定对J影响大的x, 使J小的 x_i 变化大的 x_i 收敛快小

由 $K = LQR(A, B, Q, R)$ 可得一个最佳的k

控制器: 若x可测则得到 $u=-kx$

x不可测时, 基于observer: 根据 y与u

$$\begin{cases} \dot{x} = Ax + Bu \\ y = Cx + Du \end{cases}$$ 估计x

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Luenberger Observer (龙伯格观测器)，$\hat{L}$ 工具箱里的

$$\begin{cases} \dot{\hat{x}} = A\hat{x} + Bu + L(y-\hat{y}) \\ \hat{y} = C\hat{x} + Du \end{cases}$$

object: Find L

$\hat{x}$ 为估计值，$\hat{y}$ 为估计的输出

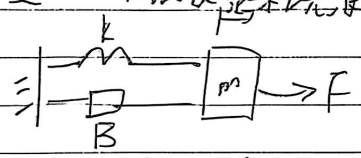
则 $\dot{\hat{x}} = (A-LC)\hat{x} + (B-LD)u + Ly$

$\dot{x} - \dot{\hat{x}} = (A-LC)(x-\hat{x})$

$= \dot{e}_x$

error: 估计与实际值的误差

建立一个新的反馈系统，使得 $e_x \to 0$



$z_1 = x$ 位置 可得

$z_2 = \dot{x}$ 速度 不可测

$y = z_1$ 位置 可测

则令 $L = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}$ 得 $A-LC$ 使 $A-LC$ 的 $\lambda_i < 0$

则可得 $\dot{\hat{x}} = (A-LC)\hat{x} + (B-LD)u + Ly$

观测器也提供了一种不需测量仅需输入输出量的方法

$A_{n\times n} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \\ \cdots & \cdots & \end{bmatrix}$，$\sum \lambda_i = tr(A)$

$\prod \lambda_i = |A|$

如果一个系统可观测则 $O \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$ $Rank(O) = n$

由 Observer 引导 u 二阶

则 $\dot{x} = (A-BK)x + BKe$

$\begin{bmatrix} \dot{e}_x \\ \dot{x} \end{bmatrix} = M \begin{bmatrix} e_x \\ x \end{bmatrix}$

M 的特征值即 $A-LC$ 与 $A-BK$ 的特征值

即 Separation principle

Observer faster than controller

LQR - Follow a desired path

If want $x_d = 5°$，$e = x_{id} - x_i$

目标 $e \to 0$，$t \to \infty$

则 $\dot{\begin{bmatrix} e \\ x_2 \end{bmatrix}} = A \begin{bmatrix} e \\ x_2 \end{bmatrix} + Bu + C$

(又转回一阶倒立摆)

令 $u = -k\begin{bmatrix} e \\ x_2 \end{bmatrix} + C_2 \Rightarrow \begin{bmatrix} \dot{e} \\ \dot{x}_2 \end{bmatrix} = A_{cl}\begin{bmatrix} e \\ x_2 \end{bmatrix}$

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PD, PSD, NPD, NSD

P: Positive, D: Definite S: Semi

有 $\begin{bmatrix} f_1 \\ f_2 \end{bmatrix}$ = ~~f~~ f 为 function

$f_1 = \dot{x}_1 = ax_1$, $f_2 = \dot{x}_2 = bx_2 + \cos x_1$

设 ~~$V(x_1)$~~ $V = x_1^2 + x_2^2$

则 $\dot{V} = \nabla V \cdot f(x) = L_f \cdot V(x)$ 即 Lie Derivative

$= \begin{bmatrix} \frac{\partial V}{\partial x_1} & \frac{\partial V}{\partial x_2} \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}$ 李导数

V: 能量, $\dot{V}$: 能量随时间变化

Lyapunov stable 仅代表① x_1, x_2 有界

② 不代表, $x_1, x_2 \to 0$

La Salle's Theorem

asymptotically stable:

① ~~$V(x) > 0, \forall$~~ $\forall x, V(x) > 0$

② $\forall x, \dot{V}(x) \le 0$

③ $\dot{V}(x) \ne 0$ 除非 $x = 0$

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